



Instructor: Date:

Learner: Regional Branch:

Program: Center:

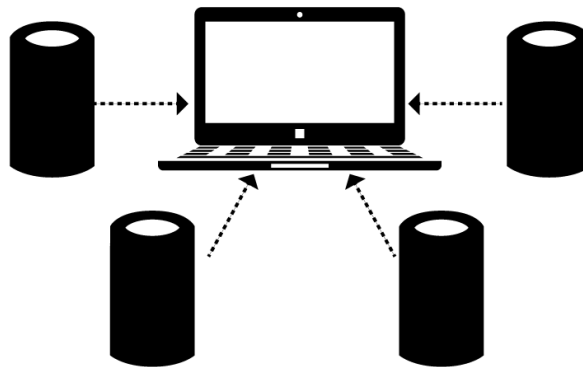
CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Another Component of Database



Learning Outcome:

Exchange written and spoken information on principles, concepts, data base objects related to query analysis, and data base engines. Describe relational database, its concepts, and principles in spoken or written texts.

Can Do Statements:

- I can talk about the importance of logical database structures.
- I can list names of logical database structures.
- I can talk about characteristics of different logical database structures.

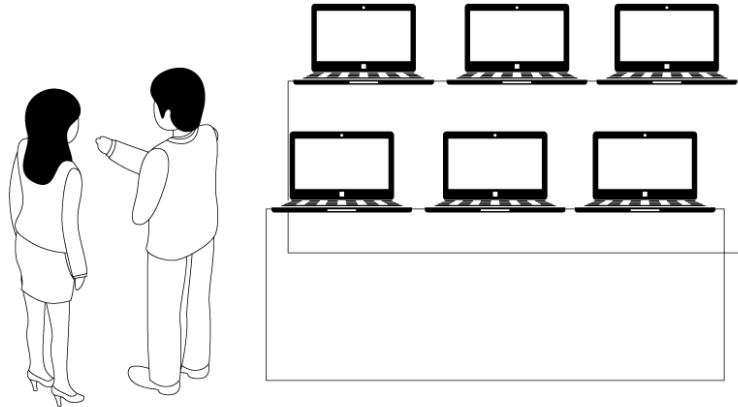
Ask your partner:

- How do you to organize information in your laptop? Do you have any logical pattern?
- Why is it important that companies have and manage their own databases?



Let's Explore!

An engineer and his assistant talk about the logical database structures.



Instructions: Read the text below.

Engineer: You are supposed to be acquainted with the physical database structures; it is time to review the logical database structures.

Assistant: The physical structures refer to the type of files and they are viewable at the operating system level, right?

Engineer: Exactly! The logical structures, on the other hand, are very important because they allocate the space for all data; they allow a fine-grained control of datafiles in the disk space. They are supposed to help you organize all the files so you can access them easily.

Assistant: The data blocks, as the smallest of the logical structures, have the finest level of granularity, don't they?

Engineer: That's right. All files are stored in them, a data block corresponds to a specific number of bytes, and you can personalize four different sizes of data blocks according to your needs. Then, we have the extent which is a specific number of data blocks that contain the same type of information.

Assistant: Above the extent, we have the segment. A segment is a set of extents.

Engineer: Correct! A segment can be temporary, and if an extent is full, another one is created into the same segment. The segments are very useful to allocate information. The last structure is a tablespace, we saw it with the physical structures. They contain and group related data blocks, extent and segments. They simplify administrative functions. Besides, they can have small or extra-large files.

Assistant: Are tablespaces supposed to be always online?

Engineer: They are usually online, but they can be set offline to avoid the access to a portion of the database, or they can make the tablespace read-only. This means you cannot change any of the data; this avoids the need for a system recovery.

Assistant: I guess I'm not supposed to forget any of this, it's all very important.





Let's Practice!

Logical Database structures are important!

Instructions: Circle the correct option to complete the sentences.

1. The logical structures, (allocate / allocates) the space for all data.
2. Logical structures (allow / allowing) a fine-grained control of datafiles in the disk space.
3. Logical structures (has / have) different structures.
4. The structures (help / helps) you to organize all the files so you can access them easily.
5. You (personalize / personalizes) logical database structures in four different sizes of data blocks according to your needs.



Let's Practice!

Take into account their characteristics!

Instructions: Circle the correct answer in order to complete the sentence.

1. The structures, from the smallest to the biggest are _____:
 - a. data blocks, extents, tablespaces and segments.
 - b. data blocks, extents, segments and tablespaces.
 - c. extents, data blocks, segments and tablespaces.
2. A data block _____ a specific number of bytes.
 - a. corresponds to
 - b. correspond to
 - c. corresponding to
3. A segment can be temporary, and if an _____ full, another one is created into the same segment.
 - a. extent is
 - b. extent are
 - c. extent isn't
4. A tablespace _____ related data blocks, extent and segments.
 - a. expands and separates
 - b. eliminates and duplicates
 - c. contains and groups





Extension Activity

Writing about what you learned.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about Logical Database Structures.
- Describe the extra information you found.
- Write a 200-250 words text talking about logical structures and their characteristics using the grammar and vocabulary studied.
- Share your work with your partners.

Self-Assessment

Yes

No

- I deepen my knowledge on database structures.
- I express information in a paragraph.
- I exchange information with my classmates.

☐	☐
☐	☐
☐	☐



Target Words

- **allocate** (verb): to distribute or share something
e.g. The logical structures allocate the space for all data.
- **acquainted** (adjective): to be familiar with
e.g. You are supposed to be acquainted with the physical database structures.
- **read only** (adjective): a file that cannot be modified
e.g. They can make the tablespace read-only. This means you cannot change any of the data; this avoids the need for a system recovery.
- **fine grained** (adjective): very specific and detailed
e.g. They allow a fine-grained control of datafiles in the disk space.
- **personalize** (verb): to customize something to a particular way
e.g. You can personalize four different sizes of data blocks according to your needs.
- **granularity** (noun): the level of detail and specificity something has
e.g. The data blocks have the finest level of granularity.



Learning Points

Be Supposed To in Affirmative and Negative Statements

Be supposed to is a grammar structure used when we want to say that someone is expected to do something, or something is expected to happen.

Be supposed to is used when you expect something to happen, or someone to do something.

It follows the **present simple** structure:

Subject + be + supposed to + verb in base form + complement

Example

*You **are supposed to** go travel tonight. (someone expects him to do it)*

In negative sentences, we use:

Subject + be not + supposed to + verb + complement

Example:

*Michelle **is not expected to** come home early.
(People expect her to arrive later)*

Examples

- *You are supposed to be acquainted with the physical database structures.*
- *I guess I'm not supposed to forget any of this, it's all very important.*



Learning Points

Understanding Data Files and Tablespaces

To understand **tablespaces**, first, we need to know that all data are stored logically in **tablespaces**. Inside the **tablespace**, we have all datafiles which are connected to a corresponding tablespace.

See the following chart:

